Pricing and Hedging of Derivatives

quantitative models, implied-volatility calibration, and numerical validation

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Abstract

This personal research project studies derivative pricing, hedging, and model calibration from a quantitative-finance perspective in which absence of arbitrage provides the theoretical foundation and numerical validation provides the operational test. The argument starts from the binomial model and the replicating portfolio, passes through risk-neutral pricing, and reaches the continuous-time limit of Black–Scholes–Merton, which remains the benchmark analytical framework for European options and for the interpretation of local risk sensitivities.

On this foundation, the project examines the limits of constant-volatility models. Discrete hedging, operational frictions, and the observed geometry of implied-volatility surfaces show that the classical benchmark, while indispensable, does not exhaust the quantitative problem. For that reason, the analysis introduces Monte Carlo simulation, hedging diagnostics, and implied-surface reconstruction as necessary steps for evaluating the empirical resilience of theoretical assumptions. The Heston model is then considered as a natural extension linking pricing, volatility dynamics, and calibration to market data.

The software infrastructure used in the project is not treated as the primary object of inquiry. Its role is experimental: it enables benchmark execution, simulation reproducibility, calibration tracking, and graphical diagnostics. The numerical evidence supports a balanced conclusion. Classical models remain central as conceptual scaffolding and as analytical benchmarks, but they encounter clear limits in the description of implied surfaces and discrete-hedging risk. Richer stochastic specifications, combined with explicit numerical discipline, produce an interpretation that is more faithful to market practice, while simultaneously requiring greater care in the reading of calibrated parameters, residual patterns, and model error.

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Chapter 1

Introduction

Quantitative derivatives finance arises from the need to assign coherent prices to contracts whose cash flows depend nonlinearly on the evolution of underlying assets, interest rates, volatility, or richer state variables. In the language of modern asset-pricing theory, the starting point is not a narrative forecast of the future but the principle of absence of arbitrage: if two portfolios generate the same terminal payoff in every state of the world, they must have the same initial cost. From this apparently elementary statement follow both risk-neutral pricing and dynamic replication. The classical literature showed that the discrete-time binomial model makes this mechanism transparent, while the continuous-time limit leads to the Black–Scholes–Merton structure, which remains the benchmark paradigm for the analysis of European options [1, 2, 3].

Over time, however, it became equally clear that the strength of the classical paradigm is inseparable from its limits. Constant volatility, continuous hedging, and lognormal spot dynamics are extraordinarily powerful analytical devices, yet they are difficult to reconcile fully with market data and with the operational microstructure faced by actual trading desks. Implied-volatility surfaces exhibit smiles, skews, and nontrivial term structures that cannot be summarized satisfactorily by a single constant volatility parameter. Hedging is implemented in discrete time, under transaction costs, jump risk, finite sensitivity estimates, and explicit dependence on model choice. In other words, the closed-form formula does not close the problem; it reorganizes it into a highly useful theoretical benchmark while leaving open the question of empirical validation and numerical robustness.

This project is positioned precisely in that space. Its purpose is not to present a software platform as an end in itself, but to study a quantitative problem: to what extent the main pricing and hedging models for options describe observed prices, implied-volatility geometry, and the hedging error that appears once replication is implemented discretely. The software is used as methodological infrastructure for benchmarks, simulation, and calibration, not as the main subject of the report.

1.1 Context and motivation

Derivatives provide a privileged laboratory for studying the coherence between economic theory, stochastic probability, and numerical methods. Relative to many other areas of finance, the mapping between modelling assumptions and economic value is unusually direct: a change in the dynamics of the underlying, in the treatment of volatility, or in the rebalancing frequency is reflected immediately in contract prices and in the risk profile of the hedge portfolio. This makes the field particularly suited to an investigation that does not merely present formulas, but instead connects analytical results, numerical comparison, and observable empirical phenomena.

The structure of the report is designed around that objective. The conceptual backbone is grounded in absence of arbitrage, replication, Black–Scholes–Merton, Greeks, Monte Carlo methods, and implied-volatility calibration. The computational material is retained only insofar as it is needed to guarantee reproducibility, methodological transparency, and experimental correctness. Theory and the interpretation of results therefore remain the core of the document.

This choice is also disciplinary. In a serious quantitative-finance project, the main intellectual interest does not lie in the bare existence of a toolkit, but in the quality of the analysis that the toolkit makes possible. That means prioritising a set of fundamental questions: in what sense is risk-neutral pricing still the reference point for modern models? Which limits emerge when theoretically continuous hedging is implemented on discrete trading dates? Why does the implied surface constitute strong empirical evidence against constant-volatility descriptions? To what extent does a stochastic-volatility model such as Heston improve the representation of observed prices, and at what cost in terms of numerical complexity, identifiability, and stability of calibration? These are the questions that organise the report.

1.2 Research question and central claim

The main research question can be stated as follows:

> *How effective are classical and stochastic models in option pricing and hedging, and how can a dedicated quantitative toolkit support their calibration, comparison, and numerical validation?*

This question contains three layers that will be treated jointly. The first is theoretical: the arbitrage-free foundation of pricing and the relation between the risk-neutral measure and the replicating portfolio must be made explicit. The second is empirical: one must verify how implied surfaces and hedging experiments reveal the limits of simpler models. The third is

methodological: one must show how a dedicated numerical infrastructure enables systematic comparison across approaches while preserving input traceability, run reproducibility, and clarity of assumptions.

The central claim defended in this report is the following:

> *Classical models provide an indispensable theoretical basis for understanding derivative pricing and hedging, but they display significant limitations in the description of implied-volatility surfaces and discrete-hedging risk; richer models, combined with systematic numerical validation, produce an analysis that is materially closer to market practice.*

This claim does not imply the obsolescence of the classical paradigm. Rather, it assigns it its correct place. The Black–Scholes–Merton model remains the conceptual and analytical benchmark against which deviations, discretisation error, and the need for richer specifications are interpreted. At the same time, numerical evidence and implied surfaces suggest that, for many operational questions, constant volatility is not an adequate approximation. The real problem is therefore not to choose between pure theory and empirical practice, but to understand how the two constrain one another.

1.3 Contributions of the project

The main contributions of the project can be summarized in four points.

1. It provides a coherent reconstruction of the theoretical framework for derivative pricing and hedging, linking the binomial model, risk-neutral pricing, Black–Scholes–Merton, and the analysis of sensitivities.
2. It discusses the limits of constant-volatility models in the light of discrete hedging, Monte Carlo simulation, and the evidence conveyed by implied-volatility surfaces.
3. It develops a numerical comparison and calibration study with particular attention to the Heston model, error metrics, and residual diagnostics.
4. It documents the use of a quantitative toolkit as experimental support for reproducible benchmarks, implementation validation, and the connection between theory and computational evidence.

The first three contributions belong directly to quantitative finance; the fourth provides the methodological support that makes them experimentally credible. This hierarchy is intentional and governs the composition of the document.

1.4 Methodological framework

The report is organised around three methodological blocks. The first develops the theoretical structure of arbitrage-free pricing, replication, stochastic calculus, and the principal sensitivities used in derivatives analysis. The second introduces the numerical workflows required for pricing, simulation, and calibration. The third interprets the empirical outputs generated by those workflows, with particular attention to implied-volatility surfaces, residual diagnostics, and the calibration of the Heston model. Notation is standardised throughout so that theoretical statements, numerical experiments, and graphical diagnostics remain mutually consistent.

Operationally, the methodology is threefold. First, the report develops a theoretical reconstruction based on the classical literature on arbitrage, replication, Black–Scholes–Merton, Monte Carlo methods, and stochastic volatility [4, 5, 2, 3, 6, 7]. Second, it uses numerical benchmarks and reproducible workflows to verify consistency between known formulas, simulation output, and calibration procedures [8, 9]. Third, it interprets results not only at the level of prices but also at the level of financial meaning: sensitivities, hedging error, strike–maturity structure, residual geometry, and parameter stability.

A further methodological point deserves emphasis. Whenever a quantity can be represented both analytically and numerically, the project treats the analytical representation as the benchmark and the numerical representation as the test of robustness. Whenever only a numerical procedure is available, the project decomposes the reported output into model risk, discretisation error, and statistical uncertainty. This decomposition is particularly important for Monte Carlo estimates and for calibration objectives, where an apparently precise number can conceal substantial numerical instability.

1.5 Structure of the report

The report is organised into six main chapters. Chapter 2 reconstructs the arbitrage-free framework from which risk-neutral pricing, the binomial model, the continuous-time limit, and the Greeks emerge. Chapter 3 addresses the operational problem of pricing and numerical hedging, with emphasis on discrete rebalancing, Monte Carlo simulation, and the limits of the constant-volatility paradigm. Chapter 4 moves from implied-volatility geometry to Heston calibration, clarifying the link between theory and market data. Chapter 5 gathers the main benchmarks, experiments, and empirical interpretations. Chapter 6 formulates the final answer to the research question and discusses limits and extensions. The appendices collect implementation support and supplementary analyses that strengthen the main body without disturbing its theoretical balance.

1.6 Positioning of the software

It is useful to clarify the role of the software from the outset in order to avoid a common ambiguity in computational finance projects. In this report, the toolkit is not introduced as the central object of design, nor as a product to be documented for its own sake. Its role is that of an experimental environment in which quantitative-finance models can be implemented, tested, and validated. This distinction has concrete consequences for the writing: the report does not begin with the repository, does not organise its table of contents around commands, and does not attribute theoretical centrality to architectural details that are merely instrumental.

This choice does not diminish the importance of implementation quality. On the contrary, it makes that importance more precise. A toolkit used to compare pricing models, reconstruct implied surfaces, or measure hedging error must guarantee at least a separation between orchestration and numerical engines, explicit control of inputs, regression tests, seed reproducibility, and parameter traceability. Software enters the argument not as an end but as a condition for the reliability of numerical evidence. The project therefore adopts a simple criterion: whenever an implementation choice improves transparency or the credibility of results, it deserves discussion; when a technical detail does not add information relevant to the financial problem, it is compressed or moved to an appendix.

Chapter 2

Theoretical framework

This chapter reconstructs the theoretical backbone of the project. The objective is not merely to restate known results, but to show how they form a continuous logical chain: from absence of arbitrage to the equivalent martingale measure, from the binomial model to risk-neutral valuation, from discrete replication to the continuous-time limit of Black–Scholes–Merton, and finally to the main sensitivities used in risk management. The central idea is that the price of a derivative should not be interpreted as a subjective forecast of the future value of the underlying, but as the coherent cost of replication or, more generally, as the value compatible with the no-arbitrage structure of the market [10].

2.1 Absence of arbitrage and the fundamental theorem of asset pricing

Consider a discrete-time market with a risk-free asset B_t and one or more risky assets. Intuitively, an arbitrage opportunity exists when one can construct a self-financing portfolio with non-positive initial cost and terminal payoff that is almost surely non-negative and strictly positive with positive probability. Excluding such opportunities is the minimal economic consistency condition for a market model. In modern theory, this requirement is not isolated; it is linked to a much stronger probabilistic property, namely the existence of a measure under which discounted asset prices become martingales [4, 5].

Definition 2.1. In a finite-horizon market, *absence of arbitrage* holds if there exists no self-financing strategy ϕ such that $V_0(\phi) \leq 0$, $V_T(\phi) \geq 0$ almost surely, and $\mathbb{P}[V_T(\phi) > 0] > 0$.

The fundamental theorem of asset pricing links this definition to a dual probabilistic statement. In a finite, frictionless market, absence of arbitrage is equivalent to the existence of at least one probability measure $\mathbb{Q}$ equivalent to $\mathbb{P}$ under which discounted traded prices are martingales.

If the market is complete, the measure is unique. This uniqueness matters economically because it implies that every replicable contingent claim admits a unique arbitrage-free price.

Theorem 2.2 (Discrete form of the FTAP)**.** *In a discrete-time, finite-state, frictionless market, absence of arbitrage is equivalent to the existence of at least one probability measure* $\mathbb{Q}$ *equivalent to* $\mathbb{P}$ *such that the discounted prices of traded assets are martingales under* $\mathbb{Q}$*. If the market is complete, the measure* $\mathbb{Q}$ *is unique.*

The economic meaning of the theorem is twofold. First, it legitimises risk-neutral pricing as a valuation method. Second, it clarifies that the measure $\mathbb{Q}$ should not be read as the investors' true subjective distribution, but as a pricing measure: it is the probability law under which the market discounts payoffs consistently with no arbitrage. This distinction between physical and risk-neutral probability is fundamental in derivatives finance. The former is relevant when estimating expected returns, realised volatility, or real-world risk scenarios; the latter is relevant when valuing a contingent payoff consistently with observable replicating instruments.

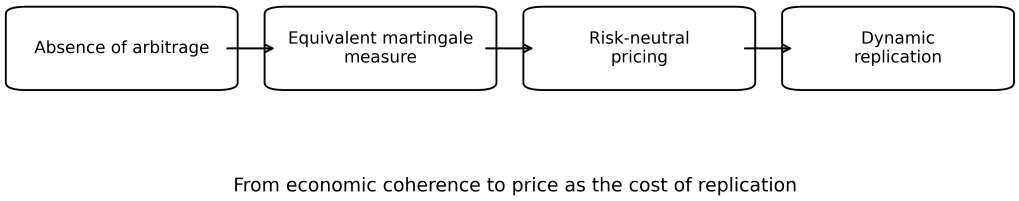


Figure 2.1: Logical chain from absence of arbitrage to risk-neutral valuation and dynamic replication.

The relation between completeness and price uniqueness already appears in the one-period case. If the number of independent traded securities is sufficient to replicate every payoff, then the derivative price is fixed by replication. If instead the market is incomplete, multiple pricing kernels can remain compatible with no arbitrage, and additional valuation criteria become necessary. Much of the later theory, including stochastic-volatility and incomplete-market approaches, can be read as a structured response to this point.

2.2 Risk-neutral pricing and the binomial model

The Cox–Ross–Rubinstein binomial model remains the most transparent environment in which the preceding logic can be observed directly [1]. Suppose that the current underlying price is S_0 and that at the next date it can take only two values: $S_u = uS_0$ and $S_d = dS_0$, with $u > d > 0$. Let $B_1 = (1 + r)B_0$ denote the value of the risk-free asset. Consider a derivative with terminal payoff C_u in the up state and C_d in the down state.

Pricing amounts to finding a portfolio formed by Δ units of the underlying and a position β in the risk-free asset such that the portfolio payoff matches the derivative payoff in both states:

$$\Delta S_u + \beta(1 + r) = C_u, \tag{2.1}$$

$$\Delta S_d + \beta(1 + r) = C_d. \tag{2.2}$$

Subtracting the equations yields the binomial hedge ratio,

$$\Delta = \frac{C_u - C_d}{S_u - S_d}, \tag{2.3}$$

while the risk-free component follows by substitution. The initial cost of the replicating portfolio is therefore

$$C_0 = \Delta S_0 + \beta. \tag{2.4}$$

This identity is already a pricing formula. It does not require preferences, utility, or subjective probabilities; it requires only the possibility of replication in an arbitrage-free market.

The same result can be written in risk-neutral form by introducing the probability

$$q = \frac{(1 + r) - d}{u - d}, \tag{2.5}$$

which satisfies $0 < q < 1$ exactly when the no-arbitrage condition $d < 1 + r < u$ holds. The derivative price can then be rewritten as

$$C_0 = \frac{1}{1 + r} \left(qC_u + (1 - q)C_d \right). \tag{2.6}$$

The current value is therefore the discounted expectation of the payoff under the risk-neutral measure. The key conceptual point is that q need not coincide with the physical probability of an upward move; it is the probability that makes the discounted underlying price a martingale.

Proposition 2.3. *In the one-period binomial model, if $d < 1 + r < u$, then the price of any replicable payoff coincides both with the cost of the replicating portfolio and with the discounted expectation of the payoff under the risk-neutral measure defined by $q = ((1 + r) - d)/(u - d)$.*

The binomial model is more than a pedagogical construction. It makes three ideas explicit that remain valid in much richer settings. First, the price of a derivative depends on the possibility of dynamic hedging, not on a narrative guess about the future. Second, the risk-neutral measure emerges from market structure, not from psychology. Third, the hedge ratio is the discrete precursor of the continuous-time delta.

2.3 From the discrete case to the continuous limit

By repeating the binomial argument on increasingly fine time grids and selecting parameters appropriately, one reaches the continuous-time limit culminating in the Black–Scholes–Merton model. Under the physical measure, the underlying price follows a geometric Brownian motion,

$$\mathrm{d}S_t = \mu S_t \,\mathrm{d}t + \sigma S_t \,\mathrm{d}W_t, \tag{2.7}$$

where μ is the historical drift and σ is the constant instantaneous volatility. Under the risk-neutral measure $\mathbb{Q}$, the drift is replaced by the risk-free rate r:

$$\mathrm{d}S_t = rS_t \,\mathrm{d}t + \sigma S_t \,\mathrm{d}W_t^{\mathbb{Q}}. \tag{2.8}$$

This substitution is not a formal trick. It is the continuous-time manifestation of no arbitrage.

For a sufficiently regular derivative price $V(S,t)$, Itô's lemma gives

$$\mathrm{d}V = \left(\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + \mu S \frac{\partial V}{\partial S}\right) \mathrm{d}t + \sigma S \frac{\partial V}{\partial S} \,\mathrm{d}W_t. \tag{2.9}$$

Constructing a self-financing portfolio long the derivative and short $\Delta = \partial V/\partial S$ units of the underlying eliminates the stochastic term. In the absence of arbitrage, an instantaneously riskless portfolio must earn the risk-free rate, leading to the Black–Scholes–Merton partial differential equation:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0. \tag{2.10}$$

For a European call with payoff $(S_T - K)^+$, the closed-form solution is

$$C(S_0, 0) = S_0 N(d_1) - Ke^{-rT} N(d_2), \tag{2.11}$$

with

$$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}. \tag{2.12}$$

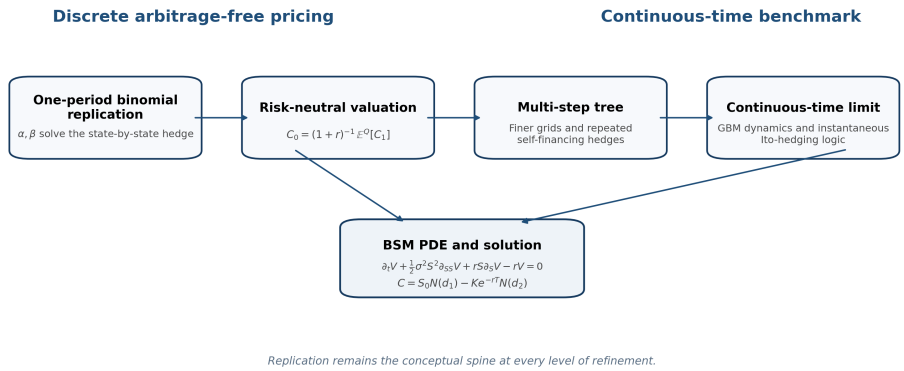


Figure 2.2: Conceptual transition from binomial replication to the Black–Scholes–Merton PDE and the closed-form solution for European options.

The deepest point is not the elegance of the formula, important as that is, but its interpretation. The Black–Scholes–Merton formula does not say that the market literally believes in lognormal spot dynamics in a naive descriptive sense. It says that if the underlying can be modelled by a GBM with constant volatility and if the derivative can be replicated continuously in a frictionless market, then the coherent option price must coincide with the solution of the preceding PDE. The formula is therefore normative and structural: it provides the benchmark against which empirical deviations and richer models are judged.

2.4 Economic interpretation of Black–Scholes–Merton

The Black–Scholes–Merton model remains central even when its assumptions are violated because it organises the economic intuition of the problem. A European call can be read as a combination of directional exposure to the underlying and risk-free financing; its price increases with spot, volatility, and time to maturity, and decreases with the strike. These dependencies, which appear in the formula as partial derivatives, correspond to structural properties of the convexity of the payoff.

One of the most important insights concerns volatility. For a convex payoff, increasing the dispersion of outcomes tends to increase the expected payoff under the pricing measure. This is why option trading is, to a large extent, trading convexity and volatility. Yet the classical formula incorporates volatility as a single constant number. In practice, market participants invert the formula to obtain the *implied volatility* that makes an observed price compatible with the model. The fact that this quantity varies systematically with strike and maturity already foreshadows the difficulties discussed later in the report.

A more quantitative way to state the same point is to observe that the Black–Scholes–

Merton formula embeds all higher-order distributional information into the single scalar σ. Once the market quotes require a surface $\sigma_{\text{impl}}(K, T)$ rather than a constant, the one-parameter representation ceases to be a complete description of the pricing map. The empirical problem is therefore not that Black–Scholes–Merton is mathematically incorrect, but that its reduced parameterisation is too rigid to represent the cross-section of observed prices.

2.5 Greeks and sensitivities

Risk sensitivities are the operational language of derivatives risk management. In the Black–Scholes–Merton setting they admit closed-form expressions, which is why the model is still used as a local hedging engine even in more complex environments. Delta,

$$\Delta = \frac{\partial V}{\partial S}, \tag{2.13}$$

measures the sensitivity of the derivative price to small spot movements. Gamma,

$$\Gamma = \frac{\partial^2 V}{\partial S^2}, \tag{2.14}$$

measures the rate of change of delta and reflects payoff convexity. Vega,

$$\mathcal{V} = \frac{\partial V}{\partial \sigma}, \tag{2.15}$$

summarises exposure to implied volatility, while theta and rho describe, respectively, time decay and interest-rate sensitivity.

Operationally, these objects are not mere partial derivatives. Delta identifies the number of underlying units needed to neutralise local directional risk. Gamma explains why a delta-neutral hedge must be updated dynamically: if the option payoff were linear in spot, one hedge would suffice; because the payoff is convex, delta itself changes with spot and the hedge must be rebalanced. Vega, in turn, makes clear that an option position contains volatility exposure even when the trader believes directional risk to have been neutralised.

In portfolio practice, first-order sensitivities are often insufficient. Vanna and volga, as well as other cross-sensitivities with respect to spot, volatility, and time, become relevant when smile moves or multi-leg structures are analysed. The theoretical framework emphasises the link between delta-hedged P&L and the difference between implied and realised volatility [10]. In continuous time and under a locally smooth model, a useful heuristic decomposition is

$$\mathrm{d}\Pi_t \approx \Theta_t \,\mathrm{d}t + \frac{1}{2}\Gamma_t S_t^2 \left(\sigma_{\text{impl}}^2 - \sigma_{\text{real}}^2(t)\right) \mathrm{d}t + \mathrm{d}M_t, \tag{2.16}$$

where $\mathrm{d}M_t$ collects residual martingale and discretisation terms. The message is that hedging does not eliminate every risk; it chooses which factors are neutralised and which remain deliberately open.

2.6 From price to hedge

The replicating portfolio is the bridge between valuation and trading strategy. In an ideal complete market, price and hedge are two faces of the same construction: the initial cost of the replicating portfolio determines the derivative price, and the composition of the portfolio indicates the strategy to be followed over time. In the binomial model this link is explicit and finite; in Black–Scholes–Merton it takes the form of continuous rebalancing of the underlying position. In both cases, however, the logic is the same: a derivative is not valued by verbal analogy with possible future scenarios, but by the trading structure required to reproduce its payoff.

This point has an essential methodological consequence for the remainder of the project. When discrete hedging, Monte Carlo simulation, implied volatility, or Heston calibration are discussed later on, the conceptual benchmark remains the framework built here. Even when markets are incomplete, valuation continues to be guided by coherence with observed prices, no-arbitrage restrictions, and the dynamics of the relevant risk factors. Extensions do not cancel the basic theory; they represent its adaptation to richer data and more realistic operational settings.

In summary, the arbitrage-free paradigm gives the project three conceptual anchors. First, pricing is a problem of coherence rather than a problem of pure prediction. Second, the risk of a derivative is not exhausted by its initial price, but must be interpreted through sensitivities and the dynamics of the hedge. Third, every numerical or stochastic extension should be judged by its ability to preserve this coherence while improving empirical realism. On that basis, it is now possible to move to the practical limits of the classical benchmark and to the tools required to address them.

Chapter 3

Numerical pricing and hedging

The classical theoretical framework becomes fully informative only when it is subjected to numerical and operational scrutiny. At this stage, the problem is no longer simply to derive a coherent arbitrage-free price, but to understand how hedging behaves in discrete time, which errors emerge from simulation, and to what extent constant volatility remains an acceptable description once one moves from the analytical plane to the experimental one. The chapter therefore treats pricing and hedging as numerical problems while keeping the theory of the previous chapter as the conceptual benchmark.

The classical pricing model remains the analytical reference point of the entire argument, but its usefulness increases when its limitations are measured precisely. This chapter addresses four issues that emerge naturally once the idealisation of continuous hedging under constant volatility is abandoned: discrete hedging, replication error, Monte Carlo simulation as a numerical laboratory, and the financial motivation for stochastic-volatility models. The guiding idea is not to abandon the arbitrage-free paradigm, but to test it under more realistic operational conditions.

3.1 Discrete hedging as an operational problem

In the Black–Scholes–Merton model, delta hedging is continuous in time. This assumption is essential for eliminating exactly the stochastic term in the dynamics of the replicating portfolio, but it is evidently unrealistic in an actual trading environment. Portfolios are rebalanced at finite intervals, prices can move discontinuously between two updates, transaction costs prevent instantaneous infinite adjustment, and parameter estimates, including volatility, are themselves uncertain. Delta hedging as observed in practice is therefore discrete, approximate, and inevitably noisy.

The difference between continuous and discrete hedging is not a mere implementation detail;

it is an economic source of risk. If a short call is hedged with Δ_t units of the underlying and the portfolio is updated only at dates $t_0 < t_1 < \cdots < t_n = T$, the terminal P&L depends not only on the initial option value and on the path of the underlying, but also on the rebalancing frequency, on realised volatility, and on the way the delta itself is estimated. In the ideal limit, the terminal hedging error tends to zero as the rebalancing mesh goes to zero. In practice, a residual remains, and that residual should be interpreted as replication error.

A useful local decomposition, often emphasised in the literature, suggests that the delta-hedged P&L of a short option can be written, to first order, as the sum of a theta contribution and a gamma contribution linked to realised variance [10, 11]. In heuristic discrete form,

$$\Delta\Pi \approx \Theta\,\Delta t - \frac{1}{2}\Gamma S^2(\Delta x)^2, \tag{3.1}$$

where Δx denotes the relative return of the underlying over the observation interval. A more informative continuous-time approximation is

$$\Pi_T \approx \frac{1}{2}\int_0^T \Gamma_t S_t^2 \left(\sigma_{\text{imp}}^2 - \sigma_{\text{real}}^2(t)\right)\,\mathrm{d}t + \varepsilon_{\Delta t}, \tag{3.2}$$

where $\varepsilon_{\Delta t}$ collects rebalancing and model-misspecification errors. When realised volatility exceeds the volatility implicitly embedded in the premium, a short-gamma position can lose money even in the absence of coding mistakes. Discrete hedging is therefore directly connected to the economics of volatility trading.

3.2 Replication error, costs, and microstructure

Replication error does not arise from a single cause. At least four factors deserve separate attention.

First, time discretisation. Even in the absence of transaction costs, a portfolio rebalanced daily or weekly cannot cancel intrastep spot movements. Because the payoff is convex, large oscillations between two hedge dates generate a P&L residual that typically grows with gamma and realised variance.

Second, transaction costs. Every rebalance requires buying or selling the underlying and therefore incurs bid–ask spreads, commissions, and, in less liquid contexts, market impact. Increasing the hedge frequency reduces discretisation risk but raises the certain cost of adjustment. Hence the classical trade-off between replication accuracy and operational cost, discussed by Avellaneda and Paras through enlarged effective volatility under transaction costs [12].

Third, model risk. If delta is computed under Black–Scholes–Merton but the underlying

follows dynamics with stochastic volatility, jumps, or heavier tails than Gaussian, the hedge is calibrated locally to a world different from the one actually realised. Even under frequent rebalancing, the residual does not vanish completely. This issue is especially relevant for the numerical experiments of this project, where the hedging engine may use Black–Scholes–Merton Greeks while paths are generated under richer specifications [8].

Fourth, input uncertainty. The implied volatility used to compute delta is not directly observed but inferred from prices. Errors in prices, sparsity of surface data, or unstable smoothing procedures feed directly into the hedge. Mature quantitative practice therefore concerns not only the correct application of a formula but also the quality of data and the robustness of the numerical infrastructure used to process them.

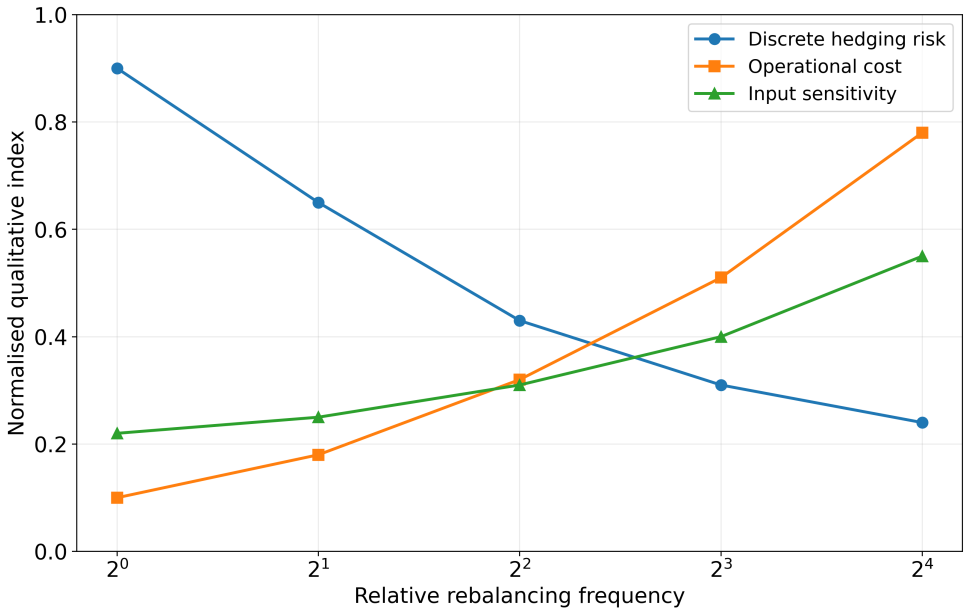


Figure 3.1: Qualitative trade-offs in discrete hedging: increasing rebalancing frequency reduces discretisation risk but raises costs, input sensitivity, and operational-control requirements.

3.3 Monte Carlo as a pricing method and risk laboratory

Monte Carlo simulation occupies a special position in derivatives theory. On the one hand, it is a pricing method: it generates paths for the underlying, computes the terminal payoff along each path, and estimates the discounted expectation under the risk-neutral measure. On the other hand, it is a numerical laboratory: it makes it possible to analyse statistical error, discretisation sensitivity, seed stability, tail dependence, and the impact of dynamic structures for which simple closed forms are unavailable [6, 13].

For a European call under GBM, the standard Monte Carlo estimator is

$$\widehat{C}_N = e^{-rT} \frac{1}{N} \sum_{i=1}^{N} (S_T^{(i)} - K)^+. \tag{3.3}$$

If the paths are independent and identically distributed under the risk-neutral measure, the law of large numbers ensures convergence of $\widehat{C}_N$ to the theoretical price, while the central limit theorem gives the rate of contraction of statistical error. In practical terms,

$$\mathrm{SE}(\widehat{C}_N) = e^{-rT} \frac{\widehat{\sigma}_{\text{payoff}}}{\sqrt{N}}, \tag{3.4}$$

so halving the standard error requires roughly quadrupling the number of paths. This scaling matters once the simulation is embedded inside a calibration loop or applied to path-dependent payoffs.

The usefulness of Monte Carlo lies not only in generality. A systematic comparison between the closed-form Black–Scholes–Merton price and a Monte Carlo estimate for a standard at-the-money case makes the distinction between theoretical benchmark and statistical variability immediately visible. As the practical material also highlights [9], a single Monte Carlo run should not be interpreted as an exact number. It should be read together with confidence intervals, seeds, and discretisation settings. While the Black–Scholes–Merton price is deterministic, the Monte Carlo output is itself a stochastic object.

3.4 Discretisation, variance, and numerical control

In continuous-time models, Monte Carlo requires a discretisation of stochastic differential equations. For GBM, the exact lognormal solution removes time-discretisation bias, but in many settings of interest, such as Heston or path-dependent payoffs, one must introduce a time-stepping scheme for the path and a monitoring convention for the payoff. This creates two distinct sources of numerical error: statistical error, linked to the finite number of simulations, and discretisation error, linked to the time grid.

The distinction is fundamental. Increasing the number of paths does not necessarily correct a systematic bias introduced by an overly coarse time grid; conversely, refining the time grid without increasing the number of paths may leave statistical uncertainty high. Numerical validation must therefore state explicitly which source of error is being controlled and how. For example, if a Heston calibration appears unstable, one must distinguish whether the instability comes from a flat objective function, from parameter non-identification, or from simulation noise in the loss evaluation itself.

The architectural material also points to implementation choices that matter for experimental

quality: use of NumPy and SciPy in numerical engines, explicit seeds in some workflows, support for Gaussian or standardised Student-t innovations in the GBM simulator, and an Euler–Maruyama scheme with full truncation for the variance process in the Heston case [8]. These choices are coherent with an academic research setting and make it possible to read Monte Carlo not only as an algorithm but as a disciplined methodology for measuring stability, bias, and convergence.

3.5 Heavy tails, stress testing, and short convexity

One important reason Monte Carlo is central to a quantitative-finance project is that it allows the Gaussian assumption to be relaxed whenever it is too restrictive. The quantitative framework explicitly discusses the use of heavy-tailed innovations as a way to stress a delta-hedged portfolio and make tail risk in short-convexity positions visible [10]. This extension is economically meaningful. A short-gamma position can look well behaved under small, continuous spot moves and yet suffer severe losses under large or abrupt moves. Stress analysis does not invalidate risk-neutral pricing; it enriches the understanding of the residual risk borne by the real hedger.

In this sense Monte Carlo should not be interpreted merely as a substitute for a missing closed form. It is an experimental environment in which several risk channels can be isolated or combined: the amplitude of spot moves, volatility persistence, spot–volatility correlation, discrete barrier monitoring, variation in term structure, and so on. This modularity explains why Monte Carlo occupies such a prominent place in the empirical stage of the project.

3.6 Why constant volatility is not enough

The best-known objection to the constant-volatility paradigm comes from observed vanilla option prices. If one inverts the Black–Scholes–Merton formula to recover the value of σ implied by each observed market price, the result is generally not constant in strike or maturity. For a given expiration, out-of-the-money and in-the-money options often display different implied volatilities; across maturities, the time dimension adds a second axis to the problem. The result is a two-dimensional implied-volatility surface rather than a single parameter.

This fact is not a cosmetic anomaly. It means that the market systematically prices features of the risk-neutral distribution that constant-volatility GBM cannot represent adequately: asymmetry, leptokurtosis, volatility persistence, negative spot–variance correlation, jump risk, or simply different term-structure effects across expiries. Implied volatility should therefore not be read as a literal estimate of future realised volatility, but as a coordinate system for a

market-consistent price map.

3.7 Toward stochastic-volatility models

Among the possible extensions, stochastic-volatility models occupy a privileged position because they preserve the arbitrage-free intuition of risk-neutral pricing while introducing an additional dynamic factor capable of generating smile and skew. The Heston model is the classical reference: the underlying follows a Brownian motion with random instantaneous variance, while the variance itself evolves as a mean-reverting CIR-type process and can be correlated with the spot Brownian motion [7]. Negative correlation makes equity-style skew possible, while stochastic variance dynamics provide a more realistic description of the maturity dimension.

Even before entering into technical details, the relevance of the model should be understood as a response to a tension between two competing goals. On the one hand, one wants to preserve enough structure for pricing, calibration, and interpretation. On the other hand, one wants to overcome the empirical insufficiency of a single constant-volatility parameter. Heston represents one of the most successful compromises: it is richer than Black–Scholes–Merton, yet still structured enough to support semi-closed pricing formulas and stable numerical workflows on moderately sized datasets.

The next chapter addresses this extension not as a purely technical alternative, but as the conceptual transition from the benchmark model to the description of implied surfaces and calibration to observed prices. The relevant question is not simply which model is more sophisticated, but which model offers the best balance between theoretical clarity, parameter interpretability, empirical fit, and numerical tractability.

Chapter 4

Implied volatility and model calibration

Implied volatility is the main point of contact between the theoretical benchmark and market evidence. If the observed surface were compatible with constant volatility, the Black–Scholes–Merton model would in many contexts be sufficient to describe prices and local sensitivities. The observation of smiles, skews, and nontrivial term structures instead makes an explicit calibration stage necessary, one in which the model is forced to confront the data. Calibration should therefore not be viewed as a secondary technical add-on; it is the bridge connecting arbitrage-free theory, model choice, and empirical descriptive quality.

Implied volatility provides the most immediate common language between pricing theory and market quotes. In principle, every option price can be converted into the value of the volatility parameter that, when inserted into the Black–Scholes–Merton formula, reproduces that price exactly. The purpose of this inversion is not to claim that the market literally believes in a constant-volatility model. Rather, it translates heterogeneous prices into a common metric that makes cross-sectional and term-structure patterns visible [14]. Smiles, skews, and surfaces are therefore representations of prices, not primitive objects separate from pricing itself.

4.1 Smile and implied surface

For fixed maturity, the function $K \mapsto \sigma_{\text{impl}}(K, T)$ is usually called a *smile* or *skew*, depending on its shape. In equity markets it is common to observe decreasing skews, with higher implied volatilities for lower strikes, consistent with greater aversion to downside risk and with negative correlation between returns and volatility. Considering several maturities, the collection of curves forms a two-dimensional surface over the strike–maturity plane.

The implied surface condenses several economic dimensions. First, it reflects the asymmetry of the market-implied risk-neutral distribution of the underlying. Second, it incorporates the term structure of implied variance, that is, the way the market prices risk across horizons. Third, it makes data inconsistencies visible: excessive oscillations, lack of smoothness, or elementary violations of static no-arbitrage. For that reason, the implied surface is not useful only for volatility traders, but for anyone seeking to assess whether a set of option quotes is internally coherent.

A correct reading of the surface nevertheless requires methodological discipline. Observed nodes have higher epistemic status than any smooth surface reconstructed by interpolation. A heatmap or smoothed mesh may be highly useful for visual intuition, but it should not be mistaken for direct market observation. The operational material from which this project draws some of its results insists, correctly, on this distinction: observed data, interpolated surface, and parametric model surface are three different objects with different informational status [9].

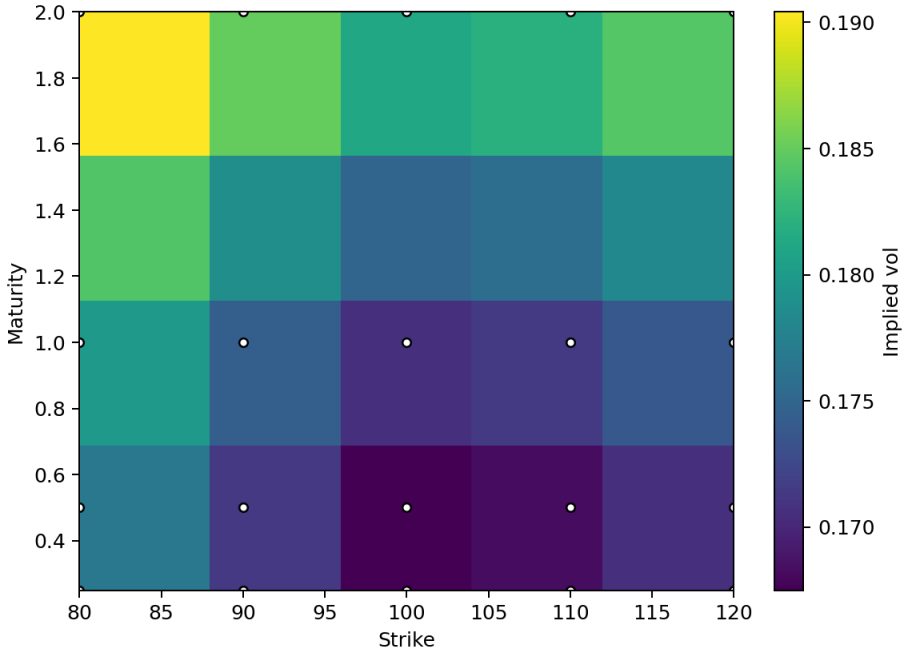


Figure 4.1: Numerical reconstruction of an observed implied-volatility surface. Market nodes must be distinguished from the smooth surface obtained by interpolation alone.

4.2 Economic interpretation of the surface

A non-flat implied surface immediately suggests that Black–Scholes–Merton does not provide a complete description of the risk-neutral distribution of the underlying. Yet the

financial meaning of surface deformations should be interpreted with care. A pronounced short-dated skew can be read as a premium demanded by the market against crash risk or downside convexity. An upward-sloping term structure of total variance may reflect risk being priced over longer horizons or incomplete mean reversion. In any case, the geometry of a figure should not be turned into a single deterministic narrative: similar surface shapes can be compatible with different combinations of leverage effect, protection demand, and balance-sheet constraints.

At a technical level, the implied surface can be linked to the risk-neutral density of terminal prices through the Breeden–Litzenberger result, according to which the second derivative of a call price with respect to strike contains state-price density information [15]. Although the present project does not reconstruct the risk-neutral density explicitly, this link clarifies why distortions of the surface should not be dismissed as visual curiosities. They correspond to a pricing distribution whose shape differs from the lognormal benchmark.

4.3 Static no-arbitrage controls

Before any parametric calibration is attempted, one should verify that the input data satisfy at least minimal static no-arbitrage conditions. For European calls without dividends, prices should be decreasing and convex in strike and increasing in maturity within appropriate bounds. In implied-volatility coordinates, these conditions do not translate into equally simple pointwise constraints, which is why many workflows compute implied volatility but keep consistency checks directly on option prices.

In the operational material, the calibration workflow appropriately separates surface reconstruction, static-arbitrage controls, and the subsequent Heston fit [8]. This separation is methodologically important. If the starting quotes already violate minimal consistency conditions, a parametric calibration can end up fitting noise or quoting errors rather than an economically plausible structure. In robust quantitative practice, calibration should never be treated as a black box that automatically “fixes” incoherent data.

4.4 The Heston model

Among stochastic-volatility models, Heston remains a widely used reference because it offers a good balance between dynamic richness and tractability [7]. Under the risk-neutral measure,

the dynamics of the underlying and the variance can be written as

$$\mathrm{d}S_t = rS_t\,\mathrm{d}t + \sqrt{v_t}S_t\,\mathrm{d}W_t^{(1)}, \tag{4.1}$$

$$\mathrm{d}v_t = \kappa(\theta - v_t)\,\mathrm{d}t + \sigma_v\sqrt{v_t}\,\mathrm{d}W_t^{(2)}, \tag{4.2}$$

with instantaneous correlation

$$\mathrm{d}\langle W^{(1)}, W^{(2)}\rangle_t = \rho\,\mathrm{d}t. \tag{4.3}$$

The parameters admit relatively clear economic interpretations: κ controls the speed of mean reversion of variance, θ the long-run variance level, σ_v the volatility of variance, ρ the correlation between price and variance shocks, and v_0 the initial variance.

The model matters for three reasons. First, negative ρ generates skews consistent with those observed in equity markets. Second, the stochastic dynamics of v_t produce non-flat implied surfaces without resorting to deterministic local volatility. Third, for European vanilla options the model admits a semi-closed representation through the characteristic function, while still leaving room for Monte Carlo implementation in path-dependent or more complex settings.

The model is not free of difficulties. The Feller condition,

$$2\kappa\theta > \sigma_v^2, \tag{4.4}$$

guarantees strict positivity of the variance process in a strong sense, but practical calibrations can approach or violate the condition without becoming useless. Moreover, different parameter vectors can generate very similar surfaces on limited strike–maturity grids, so calibration is often only partially identified. Greater flexibility therefore brings with it not only improved fit, but also a stability and interpretation problem.

4.5 Calibration as a bridge between theory and market

To calibrate a model means to choose its parameters so that model prices approach observed market prices as closely as possible. In quantitative finance, this operation has a deep meaning: it is the point at which an abstract stochastic structure is confronted with the concrete geography of market quotes. It is not enough for the model to be arbitrage-free or theoretically elegant; it must also be able to reproduce, within reasonable tolerance, the main features of the observed surface.

Formally, given market quotes $\{P_i^{\mathrm{mkt}}\}_{i=1}^n$ and a parametric model $P_i(\theta)$, calibration requires

solving an optimisation problem of the form

$$\hat{\theta} = \arg\min_{\theta \in \Theta} \sum_{i=1}^{n} w_i \, \mathcal{L}\Big(P_i(\theta), P_i^{\mathrm{mkt}}\Big) + \Pi(\theta), \tag{4.5}$$

where w_i are weights, $\mathcal{L}$ is an error metric, and $\Pi(\theta)$ may represent soft penalties over parameter regions deemed implausible or numerically unstable. In the workflow described by the architectural material, the Heston calibration minimises a price-based mean squared error enriched by soft penalties that favour numerically stable parameter regions [8]. This is reasonable in an educational and research context, even though alternative losses in implied-volatility space or vega-weighted errors are also natural.

4.6 Choice of error metric

The choice of what to minimise is not neutral. Minimising absolute price errors implicitly assigns more weight to expensive contracts, typically at-the-money and longer-dated options. Minimising implied-volatility errors produces a more homogeneous scale across the surface, but introduces the additional inversion step and can amplify numerical instability when vega is small. Relative errors can, in turn, overemphasise very cheap options. A well-designed calibration therefore depends on the economic goal of the exercise: accurate pricing in money terms, faithful smile geometry, good total-variance fit, or better subsequent hedging performance.

In this project a pragmatic view is adopted. Because the main aim is to compare the ability of models to represent prices and surfaces, metrics are read jointly: average price error, implied-volatility error, and local residual maps. This multiplicity is important because a single aggregate number can hide systematic patterns. A model can have a moderate mean squared error and yet produce the largest distortions precisely in the wings of the smile, where hedging sensitivity is often the greatest.

For later reference, it is useful to define two standard summary statistics:

$$\mathrm{RMSE}_P = \sqrt{\frac{1}{n} \sum_{i=1}^{n} \Big(P_i^{\mathrm{model}} - P_i^{\mathrm{mkt}}\Big)^2}, \tag{4.6}$$

and

$$\mathrm{RMSE}_\sigma = \sqrt{\frac{1}{n} \sum_{i=1}^{n} \Big(\sigma_{\mathrm{impl},i}^{\mathrm{model}} - \sigma_{\mathrm{impl},i}^{\mathrm{mkt}}\Big)^2}. \tag{4.7}$$

The first is more directly linked to monetary accuracy; the second is closer to the geometric interpretation of the surface.

4.7 Residual diagnostics

Residual analysis is one of the most instructive parts of the operational material and deserves to remain in the main body of the report [9]. When a market surface is compared with the surface generated by a calibrated model, it is not enough to observe that the mean error is small. One must understand where the error concentrates, whether it changes sign in the wings, whether it increases with maturity, and whether it reveals a systematic rigidity in the model.

Residual heatmaps and comparative smile slices are useful for exactly this reason. A heatmap makes the two-dimensional distribution of model-minus-market errors visible, while smile slices translate the error into a financially more transparent reading by showing whether the model underestimates or overestimates curvature at specific expiries. On the sample datasets used in the numerical part of the project, this diagnostic supports a coherent interpretation: the Heston fit returns plausible parameters and contained residuals, but does not remove every local deviation. This is exactly what one should expect from a five-parameter model calibrated on a relatively small grid: enough flexibility to capture the dominant shape of the surface, but not enough to erase every fine detail.

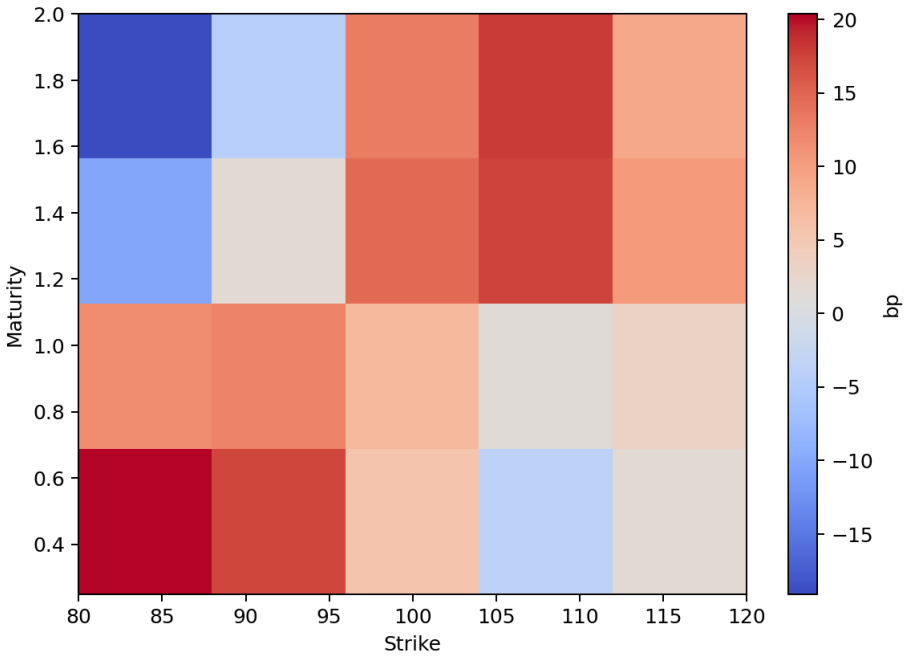


Figure 4.2: Example of local model-minus-market residual diagnostics after Heston calibration. The map reveals possible systematic structures in the fitting error.

4.8 Total variance and the reading of term structure

A particularly useful representation for comparing surfaces is total implied variance, $w(K,T) = \sigma_{\text{impl}}(K,T)^2 T$. This transformation makes the time dimension easier to read and helps disentangle strike curvature from the accumulation of variance through time. In the operational material, total-variance slices show clearly that a flatter smile does not necessarily imply a less risky market: longer maturities can exhibit lower curvature but higher total variance [9]. This point is relevant for risk management because it connects the geometry of the surface to the economic cost of protection across horizons.

4.9 Chapter summary

Implied volatility and calibration represent the decisive transition from abstract pricing theory to the empirical description of markets. The implied surface makes the limits of constant volatility observable; static no-arbitrage controls guarantee minimal consistency of the data; the Heston model provides a stochastic structure capable of generating smile and skew endogenously; calibration and residual diagnostics turn the comparison between theory and market into a well-posed quantitative problem. In the next chapter, these elements are placed within a broader numerical methodology, in which the software toolkit supports reproducibility and validation without becoming the central topic of the report.

Chapter 5

Results

This chapter presents the numerical results most relevant to the research question. The goal is not to accumulate output, but to show how benchmarks, simulations, and calibrations make it possible to compare the classical paradigm with richer extensions in a concrete way. Consistent with the overall design of the project, software appears here as an execution and experimental-control layer: tables, figures, and comparative readings remain subordinate to financial interpretation.

5.1 Pricing benchmark: Black–Scholes–Merton and Monte Carlo

The first result to establish is the correct reconstruction of the vanilla benchmark. For the standard at-the-money case $S_0 = 100$, $K = 100$, $T = 1$, $r = 0.05$, $\sigma = 0.2$, the Black–Scholes–Merton benchmark price is 10.4506 [9]. This number plays two roles in the present project. First, it certifies that the analytical engine of the toolkit is aligned with the theoretical benchmark. Second, it provides the target around which the convergence of Monte Carlo estimates should be interpreted.

Applying Monte Carlo simulation to the same case produces estimates that converge progressively to the benchmark as the number of paths increases. This is theoretically expected, but it has strong methodological value because it reminds the reader that every Monte Carlo output is an estimate with statistical error. The relevant point is not merely that the sample mean approaches 10.4506, but that the width of the confidence interval shrinks at the theoretically expected $N^{-1/2}$ rate. Figure 5.1 visualises this behaviour through a sequence of synthetic runs aligned with the information reported in the practical material.

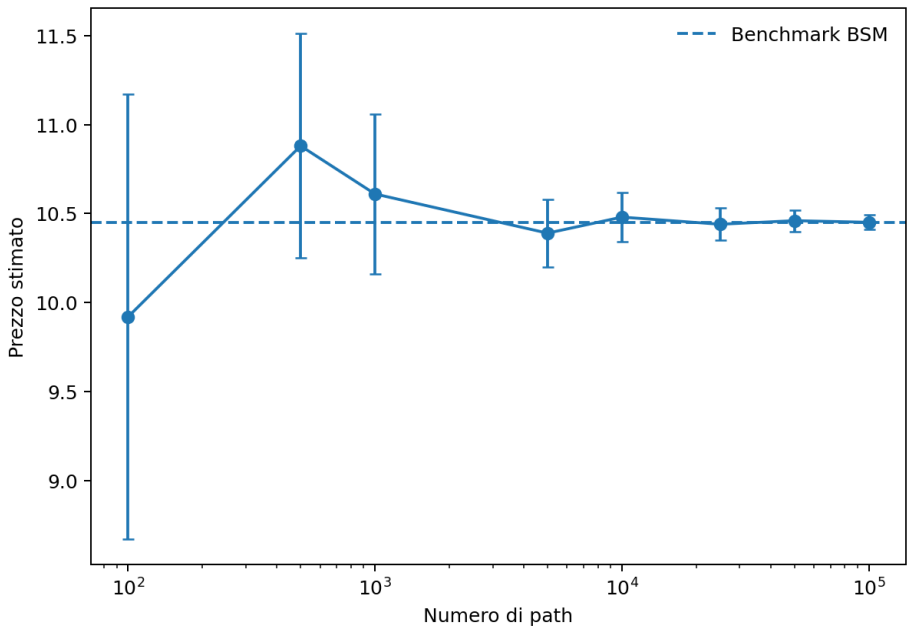


Figure 5.1: Convergence of the Monte Carlo estimate toward the Black–Scholes–Merton benchmark for the standard at-the-money case. The horizontal line marks the theoretical price 10.4506.

Two observations follow. First, the Monte Carlo engine is useful even when a closed form exists, because it validates the numerical workflow and makes estimation error explicit. Second, the comparison between the two methods shows that the toolkit is useful not only for producing a number, but also for teaching and verifying the distinction between an analytical solution and a simulated estimator. This distinction becomes crucial in later problems, where the closed form is no longer available or no longer empirically adequate.

5.2 Discrete-hedging experiments

The results on discrete hedging are especially important because they show operationally where the limit of continuous replication lies. In the classical framework, a short-option position dynamically hedged with the Black–Scholes–Merton delta should generate a P&L converging to zero as the rebalancing frequency increases. In practical simulation, what one observes is a distribution of residual P&L whose dispersion depends on rebalancing frequency, realised volatility, path structure, and payoff characteristics.

Even without presenting an excessive battery of tables, the qualitative conclusion is clear. When rebalancing is relatively frequent and the dynamics of the underlying remain close to the model assumptions, the average residual tends to be small, yet non-negligible dispersion remains because updates occur at discrete dates and paths are noisy. If instead the dynamics

depart from constant-volatility GBM, for instance by introducing heavy tails or stochastic volatility, the residuals become more variable and can exhibit asymmetric tails. This confirms the claim discussed in the theoretical chapters: the continuous-time theoretical price is a benchmark, whereas the P&L of discrete hedging is the place where the frictions of practice become visible.

A particularly informative reading concerns short-convexity positions. When the underlying moves sharply, the need to buy after rallies and sell after declines turns negative gamma into a mechanism of P&L erosion. Simulation makes this effect immediately visible, whereas the pricing formula by itself tends to conceal it. Discrete-hedging results are therefore not a technical appendix to pricing; they are an essential part of its economic meaning, because they show the operational cost of convexity and the role played by realised volatility relative to the volatility priced into the initial premium.

A compact quantitative summary is useful. If one denotes by $\Pi_T^{(m)}$ the terminal hedging P&L under rebalancing frequency m, the object of interest is not only $\mathbb{E}[\Pi_T^{(m)}]$, but also

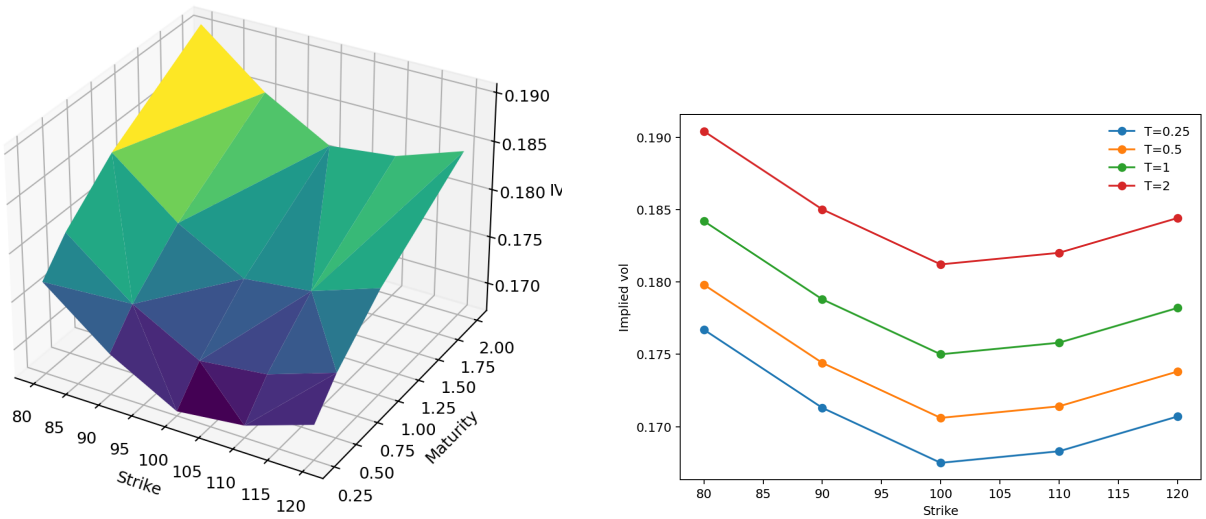
$$\mathrm{Std}(\Pi_T^{(m)}) \quad \text{and} \quad \mathrm{VaR}_\alpha(\Pi_T^{(m)}), \tag{5.1}$$

because frequency changes can affect the centre and the tails of the distribution differently. A model that improves mean hedging error while worsening tail risk is not unambiguously better in operational terms.

5.3 Reconstruction of the observed implied surface

The dataset `quotes.csv` used in the practical material contains a limited number of quotes across a small set of strikes and two maturities, yet it is sufficient to demonstrate the full workflow of implied-surface reconstruction [9]. The scientific interest of the exercise does not lie in dataset size, but in the clarity with which three informational layers are separated: observed nodes, numerical smoothing, and financial interpretation of the strike–maturity structure.

The triangulated mesh of the observed nodes reveals curvature consistent with the presence of smile and with a nontrivial term structure. A smoothed heatmap makes the global geography easier to read, but it does not replace the pointwise data. Smile slices by maturity are often even more informative because they separate strike curvature from time variation. Finally, total-variance sections show that a less pronounced smile does not necessarily imply less risk being priced by the market. This family of representations is particularly useful in the context of the project because it translates the limits of constant-volatility models into numerically readable images.



(a) Observed-node mesh.

(b) Smile slices by maturity.

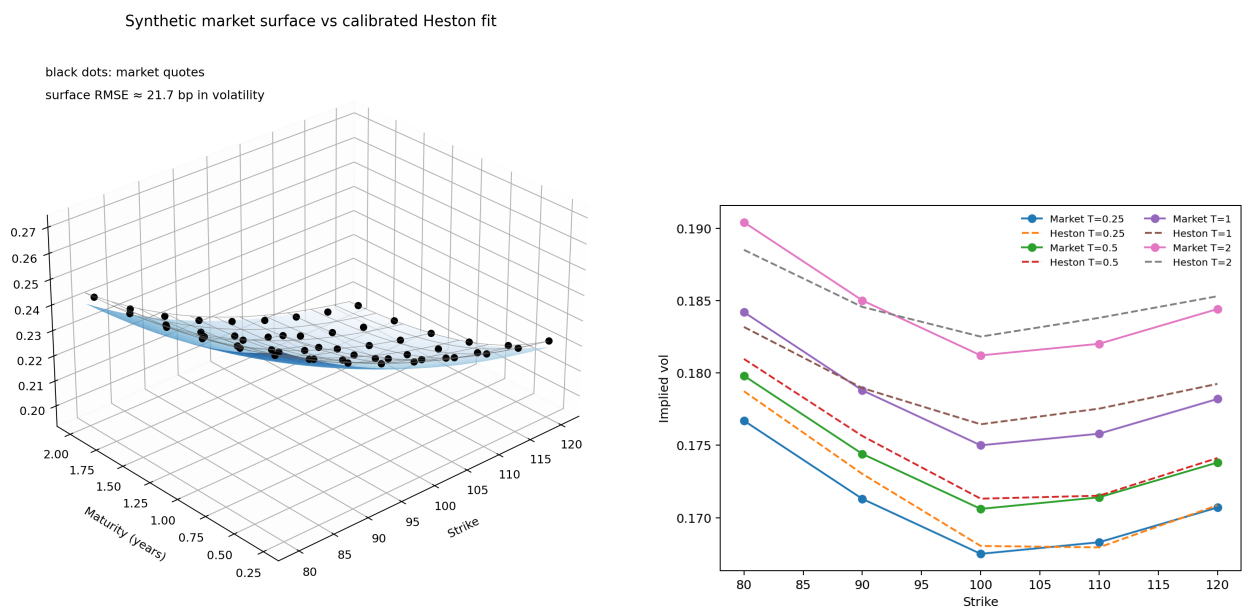
Figure 5.2: Complementary representations of the observed implied surface.

From a financial viewpoint, the most important result is not the presence of a particular geometric shape, but the fact that observed prices cannot be summarised by a single volatility parameter. The implied surface therefore makes immediately visible why the Black–Scholes–Merton benchmark cannot be regarded as a sufficient market description, even though it remains the baseline language for quoting and for many local sensitivities.

5.4 Heston calibration and fit quality

The second part of the surface experiments concerns calibration of the Heston model on the synthetic dataset `synthetic_heston_quotes.csv`. The operational material reports a calibration that returns plausible parameters, in particular $\kappa \approx 1.999$, $\theta \approx 0.0407$, $\sigma_v \approx 0.314$, $\rho \approx -0.707$, and v_0 consistent with an initial variance close to the central level of the surface [9]. These values should not be read as structural truth about the market; they show that the calibration workflow is capable of recovering a parameter configuration coherent with the sample dataset.

The comparison between synthetic market surface and the implied surface of the calibrated model shows good overall agreement. The general shape of the skew and the term structure are reproduced credibly, while small local residuals remain. This is exactly the type of result one should expect from a five-parameter model calibrated on a relatively small grid: enough flexibility to capture the main morphology of the surface, but not enough to eliminate every pointwise deviation.



(a) Market surface and fitted model surface.

(b) Comparison of smile slices.

Figure 5.3: Graphical diagnostics of the Heston fit on the synthetic dataset.

The local residuals are the most instructive part of the exercise. The model-minus-market heatmap shows that the fitting error is not distributed uniformly: there are regions of the surface in which the model slightly underestimates curvature and others in which it slightly overestimates it. The smile slices translate these differences into a more financially legible reading. For some maturities, the fit is very accurate around the money but drifts in the wings; for others, the error is more regular but persists as a small level difference. This confirms an important theoretical point: a good average error does not exhaust the evaluation of a calibration. One must always ask *how* the model is wrong.

A concise quantitative interpretation can be attached to the visual evidence. If one computes both RMSE_P and $\mathrm{RMSE}_{\sigma_{\mathrm{impl}}}$ from the calibrated surface, the first summarises monetary fit while the second summarises geometric fit. A visually acceptable 3D surface with poor wing alignment may still have a low price RMSE if most weight comes from central contracts. The two metrics therefore complement, rather than replace, graphical diagnostics.

5.5 Model comparison

Combining the previous results yields a comparison across three levels.

At the level of the *analytical benchmark*, Black–Scholes–Merton remains unmatched in transparency and speed. Whenever the vanilla assumptions are sufficient, the model offers closed forms, immediate Greeks, and an extremely powerful interpretive reference point. This

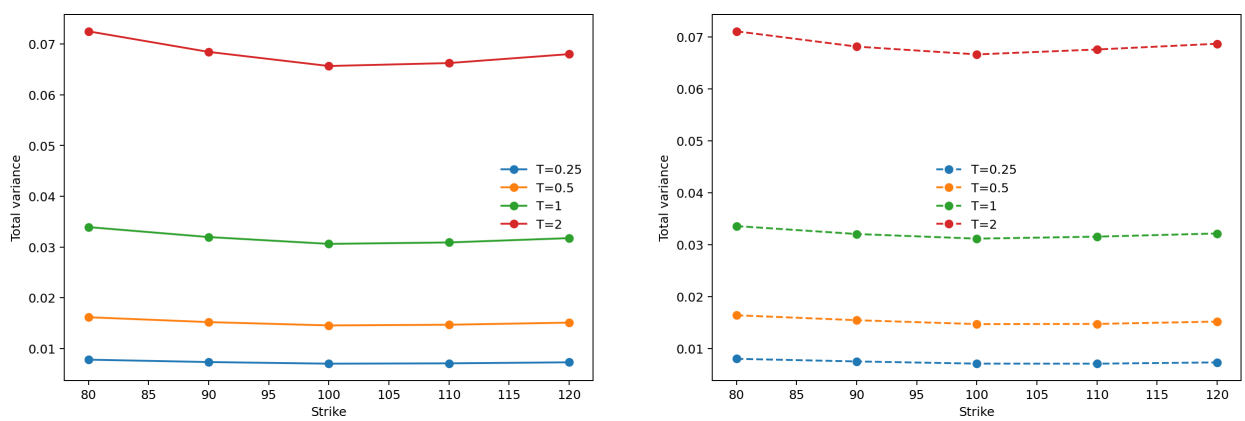
explains why the formula remains widely used in practice not only for pricing but also as a common language for quoting and risk management.

At the level of *numerical flexibility*, Monte Carlo radically expands the set of problems that can be treated. It makes it possible to handle exotic payoffs, simulate discrete hedging, introduce heavy tails, and analyse dynamics for which no closed form is available. The price paid for this flexibility is unavoidable statistical error, together with the need to control discretisation, seeds, and computational cost.

At the level of *fit to implied surfaces*, Heston represents an effective compromise. It does not attain the simplicity of Black–Scholes–Merton, but it provides a qualitatively superior description of skewed observed or synthetic surfaces. Compared with even more flexible models, it retains relatively good economic interpretability of parameters. This is why, in the present project, it is chosen as the reference model for the transition from classical theory to empirical calibration.

5.6 Results on total variance and term structure

The total-variance figures, both on the observed dataset and on the calibrated synthetic one, deserve a separate reading. Transforming implied volatility into total variance helps isolate the time effect from curvature in strike. In the observed results, longer maturities exhibit a generally higher level of total variance even when the smile appears visually flatter. In the calibrated Heston case, the model-generated total-variance surface reproduces this structure in a plausible way, while still leaving some inevitable local residuals.



(a) Observed total variance.

(b) Total variance under the calibrated model.

Figure 5.4: Qualitative comparison between observed structure and the total-variance structure implied by the calibrated model.

For the argument of the project, the important point is that total variance clarifies the

passage from the surface as a visual image to the surface as an economic object. A long-dated maturity can display a relatively flat smile while still embedding expensive protection because the market prices a larger total amount of risk over time. This observation is also relevant for hedging, because it suggests that comparing only smile curvature can be misleading if the time scale of risk is ignored.

5.7 Role of the software support in the results

Even in the experimental chapter, the conceptual hierarchy of the report should be preserved. The software is present at every stage: it executes benchmarks, generates simulations, computes implied volatility, constructs figures, and outputs calibration results. Yet the value of those results does not lie in the mere existence of a workflow. It lies in the quality of the analysis that the workflow enables. Architectural information therefore plays an ancillary, though non-zero, role. Separation between workflows and numerical engines, the presence of tests, and the file-based nature of inputs all help ensure that the results can be interpreted as quantitative evidence rather than as opaque execution artefacts.

This point is particularly clear when one compares the experiments with one another. The same infrastructure connects closed-form pricing, Monte Carlo estimation, discrete hedging, implied surfaces, and calibration diagnostics. Such experimental continuity matters because it prevents each result from being treated as an isolated episode. Instead, the reader can follow a single thread: from theoretical benchmark to numerical validation, then to the measurement of its limits, and finally to the calibration of a richer model.

5.8 Synthesis and discussion of the results

Overall, the experiments confirm the central claim of the project. The classical paradigm provides reliable and conceptually indispensable benchmarks. At the same time, discrete hedging and implied surfaces show that market practice requires richer tools at both the stochastic and numerical levels. Monte Carlo proves essential as a general method and as a laboratory for measuring error and residual risk. Heston, while not eliminating every residual, offers a substantially more faithful representation of the implied geometry of prices. The toolkit makes these comparisons reproducible, turning theoretical argument into a set of coherent empirical checks.

The numerical results acquire their full meaning only when read against the strengths and limits of the models considered.

5.8.1 Strengths of the models considered

Black–Scholes–Merton and the binomial model offer theoretical clarity, computational speed, and immediate economic interpretation. They are highly effective as benchmarks, for vanilla European pricing, and for local risk analysis through the Greeks. Their structure also makes it possible to explain rigorously the principle of absence of arbitrage and the nature of risk-neutral pricing.

Monte Carlo offers generality. It can treat complex payoffs, discrete monitoring, heavy tails, and dynamics not covered by closed-form expressions. It is not only a pricing method, but also an experimental setting in which convergence, discretisation, and statistical variability can be observed directly.

Heston represents an effective compromise between tractability and empirical realism. It introduces stochastic volatility with economically interpretable parameters, generates skew endogenously, and improves the representation of the implied surface materially. For a project that aims to remain at the boundary between theoretical rigour and numerical feasibility, this balance is especially important.

5.8.2 Limits and interpretive cautions

Alongside these strengths, the analysis also suggests several cautions. First, the fit of a model is not equivalent to structural truth. A Heston calibration that reproduces an observed surface well may still prove fragile with new data, with a different loss function, or on a larger strike–maturity grid. Calibrated parameters should therefore be read primarily as risk-neutral adaptation parameters, not as ontological descriptions of the economy.

Second, discrete hedging reminds us that the risk problem is not exhausted by matching the initial price. Two models can produce similar prices but very different hedging performance once time discretisation, costs, and misspecification are introduced. This observation also matters for calibration: good cross-sectional fit does not automatically imply good dynamic hedging properties.

Third, the numerical evidence presented here is deliberately disciplined but does not pretend to cover the full space of possible specifications. The surface datasets are sample datasets and, in the observed case, relatively small. This does not invalidate the main conclusions, which concern the direction and meaning of the results, but it does counsel care in over-interpreting any single error value or calibrated parameter vector.

5.8.3 Interpretive synthesis

Taken together, the results support a coherent picture. Derivative pricing requires an arbitrage-free theoretical structure, but that structure becomes fully informative only when confronted with the limits of discrete hedging and with the evidence contained in implied-volatility surfaces. Richer models such as Heston improve empirical adherence and descriptive capacity, but they require stricter numerical discipline and a more cautious interpretation of parameters. Software occupies the correct role of infrastructure linking these levels without replacing them. In that sense, quantitative derivatives finance appears here not as a catalogue of formulas or algorithms, but as a continuous process of comparison between theoretical coherence, empirical observation, and numerical validation.

Chapter 6

Conclusions

The research question formulated in the introduction asked how effective classical and stochastic models are in option pricing and hedging, and how a quantitative toolkit can support calibration, comparison, and numerical validation. The analysis developed in this report makes it possible to provide a structured answer.

First, classical models remain essential. The fundamental theorem of asset pricing, the binomial model, and Black–Scholes–Merton provide the conceptual structure without which derivative pricing would lose economic coherence. The language of the Greeks, the notion of dynamic replication, and the analytical benchmark for vanilla options remain indispensable tools for anyone seeking to understand the problem rigorously.

Second, that theoretical basis is not sufficient on its own to describe market practice. Discrete hedging, transaction costs and frictions, heavy-tail scenarios, and above all the non-flat geometry of implied-volatility surfaces all reveal the limits of the constant-volatility paradigm. Implied volatility therefore emerges as crucial empirical evidence: it shows where the classical benchmark remains useful and where a richer structure becomes necessary.

Third, the Heston model proves to be an especially effective extension. Although it does not eliminate every calibration residual, it generates smile and skew endogenously and produces a significantly better compromise between tractability and empirical realism. Its usefulness, however, depends on calibration quality, on the choice of loss function, and on careful reading of local residuals. A richer model is not automatically a better model unless its fit, stability, and interpretation remain under control.

Finally, the software support demonstrates its value as methodological infrastructure. The toolkit used in the project makes it possible to execute benchmarks, simulations, surface reconstructions, and calibrations in a reproducible way. The contribution of the software is not that it becomes the subject of the report, but that it makes the transition from theory to numerical evidence auditable and repeatable.

The main contributions of the project can therefore be summarised in four points: a coherent reconstruction of the theoretical framework for pricing and hedging; an analysis of the limits of constant volatility in the light of implied-surface evidence and discrete hedging; a numerical study of calibration and comparison across models; and a disciplined use of a toolkit in support of reproducible experimentation.

The project also has natural limitations. The surface datasets considered are sample datasets and do not cover the full variability of real market cases. The hedging analysis, while meaningful, does not incorporate exhaustively every possible operational friction. At the modelling level, the focus on Heston leaves open comparison with local-volatility, jump, and rough-volatility alternatives. At the numerical level, there is still room for further development in seed management, richer run metadata, and more systematic reporting.

These limits also indicate directions for future work. One natural extension would be to enlarge the empirical analysis to denser market datasets and to different classes of underlyings. A second would be to compare fit quality and hedging quality jointly across multiple horizons, in order to test whether better cross-sectional calibration translates into better dynamic protection out of sample. A third would be to explore more flexible implied-surface models while keeping economic interpretability and numerical controllability central.

In conclusion, the project supports the view that quantitative derivatives finance should be read as a field in which theory, numerics, and market evidence are inseparable. The price of an option is not a pure prediction; it is the result of an arbitrage-free structure confronted with observable data and made operational by reliable numerical tools. Classical models provide the indispensable base. Stochastic-volatility models and systematic numerical validation bring the analysis closer to market reality. The main contribution of the report lies precisely in that integration.

Appendix A

Software support, reproducibility, and implementation

The technical material related to the toolkit is placed in an appendix so that the main body of the report can remain centred on quantitative-finance arguments. This does not reduce its methodological importance: benchmarks, simulations, calibrations, and diagnostics all depend on a numerical infrastructure that is sufficiently transparent and reproducible. The sections below therefore collect the essential information needed to understand how the experiments were made executable, controllable, and repeatable.

The project uses a dedicated quantitative toolkit to perform pricing, simulation, calibration, and diagnostics. Consistent with the overall structure of the report, the software is not treated as the primary object of design, but as experimental infrastructure supporting quantitative analysis. This is more than a stylistic decision. It means that architectural aspects are discussed only insofar as they affect correctness of results, reproducibility of experiments, and transparent comparison across models.

A.1 Objectives of the toolkit in the context of the project

For the problems studied here, a useful numerical infrastructure should satisfy at least five requirements. First, it should make it possible to execute simple, verifiable benchmarks such as comparing the closed-form Black–Scholes–Merton price with a Monte Carlo estimate. Second, it should support discrete-hedging simulations and stress tests under alternative assumptions on the dynamics of the underlying. Third, it should support the construction of implied surfaces from observed quotes while preserving the distinction between raw data, smoothing, and diagnostics. Fourth, it should enable calibration of a stochastic-

volatility model and the interpretation of resulting residuals. Fifth, it should do so with enough traceability to allow experiments to be repeated with the same inputs and the same parameterisation.

These requirements matter more than the detailed naming of modules or the adoption of any particular software pattern. System structure matters because it conditions the reliability of the experimentation. From this point of view, the architectural material is useful because it shows a reasonable separation between CLI layer, application workflows, and numerical engines, sufficient for the type of analysis conducted in this report [8].

A.2 Observed general architecture

The repository used in the project adopts a terminal-first structure. A command-line interface gathers parameters and input files; application workflows translate those arguments into concrete use cases; numerical engines execute pricing, simulation, hedging, calibration, and risk optimisation; utility and I/O modules provide support functions. Methodologically, this organisation has a simple but important advantage: it separates the level of the experiment from the level of the numerical method.

For example, a pricing benchmark can be defined as a use case and invoke internally either a Black–Scholes–Merton engine or a Monte Carlo engine without relying on untracked manual procedures. Similarly, Heston calibration can be isolated in a workflow that loads quotes, sets bounds and the loss function, calls the optimisation engine, and generates interpretable output. Such separation reduces the risk of conceptual mistakes, such as confusing presentation parameters with model parameters or applying inconsistent conventions across experiments.

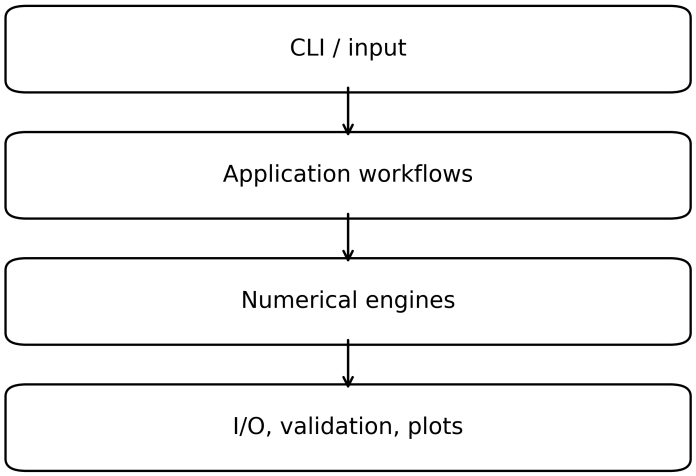


Figure A.1: Schematic view of the software support: CLI interface, application workflows, numerical engines, and supporting modules. The interest of the present project is methodological rather than architectural in the narrow sense.

A.3 Pricing, simulation, and calibration workflows

The practical material makes it possible to reconstruct a coherent operational flow. The user command selects a use case, the corresponding workflow loads data or resolves implicit parameters, and then activates the appropriate numerical engine. For vanilla pricing, the comparison between closed-form and Monte Carlo output constitutes the basic benchmark. For implied surfaces, workflows read quotes from CSV files, compute implied volatilities node by node, and generate graphical or tabular diagnostics. For Heston, the calibration workflow uses a loss function on market prices and produces estimated parameters, which can then be stored and reused [8, 9].

This organisation is also pedagogically important. It separates three moments that often remain entangled in less disciplined work: definition of the quantitative problem, execution of the numerical method, and presentation of the result. In a report such as the present one, this distinction is valuable because it keeps the financial question central: one always starts from a pricing, hedging, or calibration problem, and only then describes the numerical instrument used to address it.

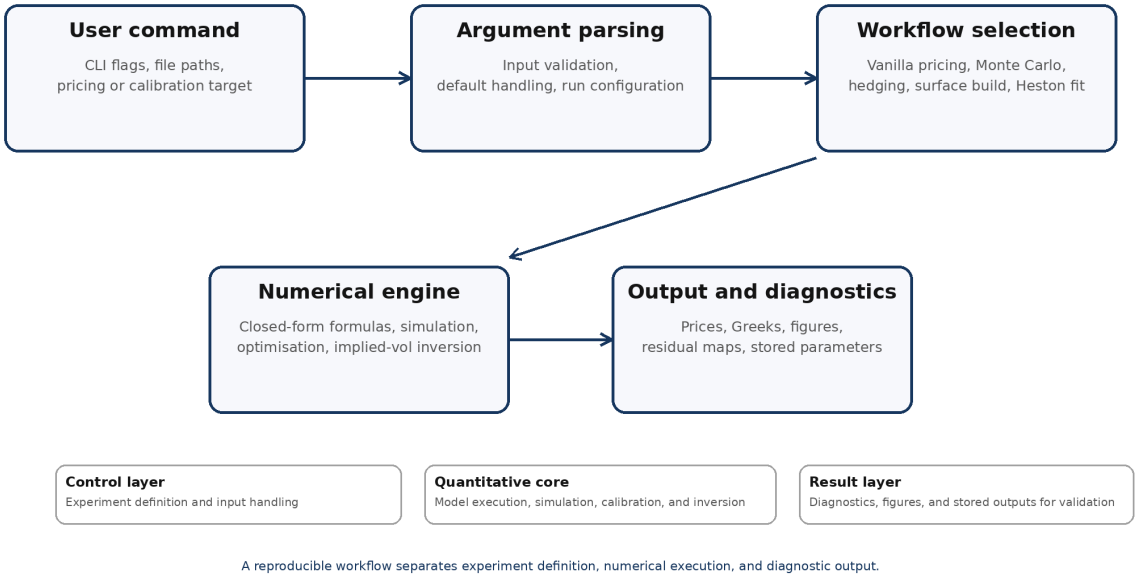


Figure A.2: Operational path reconstructed from user command to numerical result. The view is used here as support for reproducibility of experiments.

A.4 Reproducibility and implementation validation

A significant part of the value of a quantitative toolkit lies in the ability to replicate experiments. In a Monte Carlo context, declaring the seed is essential for distinguishing differences due to the method from differences due merely to simulated randomness. In calibration workflows, traceability of inputs, bounds, and the loss function is necessary for interpreting estimated parameters correctly. In implied-surface workflows, preserving original datasets and documenting smoothing procedures are equally important, because a final figure should not conceal undocumented transformations of the data.

The documented test environment reports unit and integration tests covering Black–Scholes–Merton pricing, the binomial model, discrete hedging, exotics, Heston, calibration, and some packaging or validation checks [8]. Although the report is not about software quality assurance in the narrow sense, this information is methodologically relevant: targeted tests increase confidence in the numerical support and reduce the risk that a financially interesting result is actually caused by a trivial implementation bug.

At the same time, the documented environment also indicates limits consistent with the academic character of the toolkit. The observed environment is single-process and file-based, without a fully mature CI/CD stack or advanced artefact management. These limits do not

compromise the use of the toolkit as a research laboratory for the present project, but they remind the reader that the software should be interpreted as a research instrument rather than as a production platform.

A.5 Numerical-validation criteria adopted

The project relies on four main numerical-validation criteria.

First, *consistency with closed-form benchmarks*. Whenever an analytical formula is available, as in the Black–Scholes–Merton case, it is used as the reference for interpreting Monte Carlo convergence and for assessing the plausibility of specific outputs.

Second, *robustness with respect to discretisation and seed*. Monte Carlo and Heston outputs are read together with their generation conditions, avoiding excessive precision claims for values that depend on finite sampling or temporal grids.

Third, *cross-sectional consistency*. Implied surfaces and calibrations are judged not only node by node, but also by the overall structure of errors across strike and maturity.

Fourth, *traceability of inputs*. CSV files, calibrated-parameter JSON files, model assumptions, and workflow naming conventions make it possible to reconstruct how a result was obtained.

These criteria do not replace theory; they operationalise it. An arbitrage-free model without numerical discipline can generate seemingly sophisticated but unreliable outputs. Conversely, a powerful experimental infrastructure without a theoretical framework risks degenerating into a catalogue of uninterpreted numbers. The methodology adopted in this project aims to avoid both extremes.

A.7 Minimal workflow structure

The experiments discussed in the main chapters can be grouped into four operational families:

1. **Vanilla benchmarks**: comparison between Black–Scholes–Merton price and Monte Carlo price for standard European options.
2. **Discrete hedging**: generation of spot paths, computation of delta at each hedge date, and analysis of residual P&L.
3. **Implied-surface reconstruction**: loading quotes from CSV, implied-volatility inversion, consistency checks, and graphical representation.
4. **Heston calibration**: choice of bounds, loss function, initial parameters, and production of diagnostics on surface and residuals.

A reproducible pipeline should record at least the following: input files used, model parameters, Monte Carlo seed, date and time of execution, number of paths, time step, library versions, and unambiguous names for generated output.

A.8 Numerical conventions and points of attention

To avoid interpretive ambiguity, several conventions should be stated explicitly.

- Black–Scholes–Merton benchmark prices are reported in a no-dividend setting unless otherwise stated.
- Volatilities used in benchmark examples are annualised.
- In Monte Carlo workflows, innovations may be Gaussian or standardised Student-t; in the latter case, comparison with Black–Scholes–Merton should be read as stress analysis rather than as a test of exact agreement.
- In simulation-based Heston workflows, the number of paths and the time step have a direct effect on the smoothness and stability of the loss function.
- Smooth surfaces reconstructed from a small number of observed nodes should not be over-interpreted; they serve as visual and diagnostic aids.

A.9 Expected outputs

Outputs relevant for the project include scalar prices, Greeks, simulated hedging P&L, implied-volatility tables, calibrated parameter vectors, smile and surface figures, residual heatmaps, and reusable JSON files for later experiments. In a research environment, each output should ideally be accompanied by minimal run metadata.

A.10 Why these details remain in an appendix

The placement of this material in an appendix is consistent with the general design of the report. Implementation details matter for transparency, but they should not obscure the argumentative hierarchy of the project. The main body is devoted to models, results, and financial interpretation; the operational notes collected here serve only to support the reproducibility of those results.

Appendix B

Supplementary derivations, notation, and additional analyses

This appendix gathers complementary material that supports the main body without altering its central flow: notation crosswalks, quality checklists, numerical sensitivities, conceptual ablations, and additional observations on calibrated parameters. Keeping these elements in an appendix lightens the main chapters while preserving the level of detail needed for rigorous reading and for possible future extensions of the project.

B.1 Essential notation crosswalk

In practice, one encounters slightly different conventions for spot, implied volatility, Heston parameters, and pricing output. In this report, notation is standardised to the theoretical conventions introduced in the main text. Table B.1 summarises the essential mapping.

Table B.1: Compact notation crosswalk between common shorthand labels and the notation adopted in the report.

Final notation	Observed variants	Meaning
S_t	`spot`, `S0`, `underlying`	Underlying price at time t
K	`strike`	Strike price
T	`maturity`, `tau`	Time to maturity in years
$\sigma_{\text{impl}}(K,T)$	`iv`, `impvol`	Implied volatility
v_t	`variance`, `var`	Instantaneous variance in the Heston model
$\kappa, \theta, \sigma_v, \rho, v_0$	sometimes `volvol` or `xi` for σ_v	Heston-model parameters
$\Delta, \Gamma, \mathcal{V}$	`delta`, `gamma`, `vega`	Main option sensitivities
P^{mkt}	`market price`, `quote`	Observed market price
$P(\theta)$	`model price`	Price produced by the parametric model

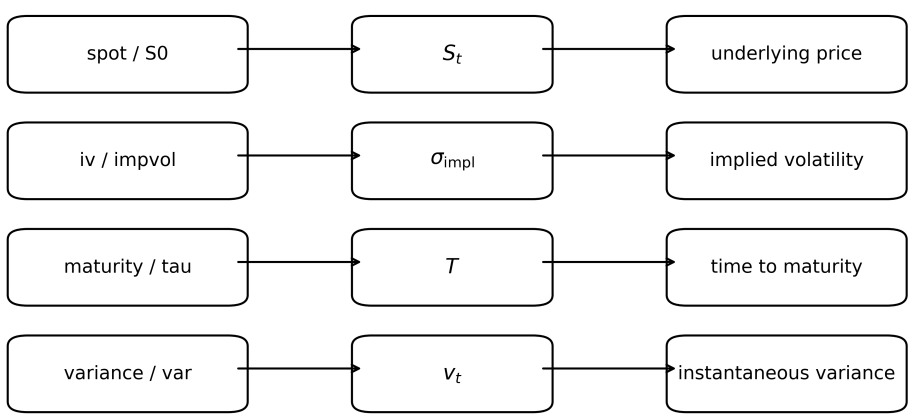


Figure B.1: Visual notation crosswalk used in the report.

B.2 Quality checklist for experiments

For every numerical experiment discussed in the report, at least the following points should be checked.

1. **Input consistency**: spot, strike, rate, maturity, and annualisation conventions should be documented unambiguously.
2. **Seed traceability**: whenever Monte Carlo is used, the seed should be explicit or recorded.
3. **Error separation**: statistical error, discretisation error, and model misspecification should be distinguished.
4. **No-arbitrage controls**: quotes used for implied-surface construction should satisfy at least elementary monotonicity and convexity conditions in price space.
5. **Fit diagnostics**: calibration should be judged not only by average error, but also by local residuals and comparative smile slices.
6. **Repeatability**: input files, parameter JSON files, and output naming conventions should make the run reconstructible.

B.3 Note on figure selection

The figures included in the report belong to two categories. Some are schematic reconstructions intended to clarify theoretical flow or methodological support; others are derived from the numerical analyses developed in the project, especially Monte Carlo benchmarks, observed implied surfaces, and Heston calibration diagnostics. In both cases their function is interpretive rather than ornamental.

B.4 Monte Carlo sensitivity to the number of paths

Convergence of Monte Carlo simulation toward the theoretical benchmark is one of the most intuitive results in the project, but that is precisely why it risks being interpreted too schematically. In practice, the number of paths N affects not only the final precision of the estimate; it interacts with the payoff type, time discretisation, and the broader procedure into which the simulation is embedded. In the pricing of a European vanilla under GBM, increasing N reduces the width of the confidence interval in a regular and predictable way. However, when simulation is used inside a Heston calibration or a multi-scenario stress test, the same increase in N can have more subtle effects.

Within calibration, each evaluation of the loss function requires repeated pricing over multiple surface nodes. If N is too small, the loss becomes noisy and insufficiently smooth, making the optimiser's task harder. The result can be an apparent instability of the model that in fact

originates in numerical noise rather than in the financial structure of the problem. Conversely, a very large N improves stability but raises computational cost, potentially making the calibration impractical. The trade-off is therefore not simply between accuracy and runtime, but between local smoothness of the loss and overall feasibility of the experiment.

A second aspect is sensitivity to the seed, especially when the number of simulations is moderate. In a purely pedagogical context, it is tempting to fix one seed and consider that sufficient. In fact, if the output changes materially across seeds, the result should not be reported as a point estimate alone but as a range of plausible estimates. A more rigorous practice is to repeat the experiment across independent seeds and to report both the mean estimate and the dispersion across runs. This becomes especially relevant when comparing models: small differences between two specifications should not be over-interpreted if they are comparable in magnitude to finite-sampling variation.

B.5 Hedging frequency as an ablation variable

Rebalancing frequency is the natural ablation variable for any discrete-hedging experiment. The main chapters show that moving from continuous to discrete hedging introduces non-removable P&L residuals. A finer reading asks how those residuals change when the number of hedge dates varies while all other parameters remain fixed.

To first order, increasing the number of hedge adjustments tends to reduce the dispersion of terminal P&L. This is theoretically consistent and unsurprising. However, the rate of improvement is not linear and depends on the volatility regime of the underlying. In relatively quiet scenarios, the move from weekly to daily hedging can be meaningful but not transformative. In heavy-tailed or volatility-shock scenarios, the marginal gain of very high frequencies can diminish because the dominant source of error is no longer only the size of the rebalancing interval, but also model misspecification and non-Gaussian intrastep moves.

Costs add a second dimension. A genuine ablation of discrete hedging should therefore be read as a two-objective problem: minimise residual risk while controlling operational cost. Without costs, the highest frequency often appears optimal. Once spreads and commissions are introduced, the problem begins to resemble an efficient-frontier optimisation. Intermediate frequencies may become economically preferable even if they do not minimise terminal-P&L variance.

B.6 Choice of loss function in calibration

One of the most underestimated sources of variability in calibration is the choice of loss function. Two calibrations run on the same dataset and the same model can produce materially different parameter vectors if one minimises price errors and the other minimises implied-volatility errors. The reason is straightforward: the two metrics do not assign the same economic weight to the same surface nodes.

When price errors are minimised, contracts with larger premiums have greater influence on the total loss. This can be reasonable if the objective is monetary accuracy on the most relevant or most traded contracts. On the other hand, that choice can tolerate visually important errors in the wings of the surface, where option prices are low but the information about the shape of the risk-neutral distribution is often especially interesting. Minimising implied-volatility errors partially reverses the weighting: the optimiser focuses more directly on the shape of the smile, but it may underweight economically important price differences when vega varies substantially across nodes.

The general methodological lesson is that no loss function is universally best. The appropriate loss depends on the intended use of the calibrated model. If the final objective is the production of a smooth descriptive surface, an implied-volatility metric can be natural. If the goal is portfolio pricing, a price-weighted or sensitivity-weighted metric may be more appropriate. In a project such as the present one, which aims at structural comparison across models, it is reasonable to keep several metrics in parallel and to interpret them jointly.

B.7 Data density and the risk of over-interpretation

The surface datasets used in the project are deliberately compact and mainly serve to make the workflow numerically readable. For that very reason, they require interpretive care. A surface reconstructed from only a few nodes can appear visually rich after interpolation, but the effective information content remains limited by the density of observed data. This is a classic source of over-interpretation in visual analysis.

The risk takes two forms. The first is to assign financial meaning to local curvatures that are actually produced by the smoothing procedure. The second is to conclude that a parametric model has “explained” the surface when the dataset is too sparse to distinguish seriously between competing specifications. In both cases the problem is not the model itself, but the insufficiency of available information.

To limit this risk, the main text preserves a strict distinction between observed nodes, smoothed heatmaps, and parametric surfaces. A useful future extension would be to add cross-validation procedures on subsets of nodes, testing whether a calibration that appears

stable on one subset preserves predictive quality on held-out nodes.

B.8 Interpretation of Heston parameters

Heston parameters have an important advantage relative to those of more flexible but less structured models: they admit direct economic interpretation. κ measures the speed of mean reversion of variance, θ the long-run variance level, σ_v the volatility of variance, ρ the leverage effect, and v_0 the initial variance. Yet interpretability does not imply perfect identification.

On small or moderately noisy datasets, different parameter combinations can produce very similar surfaces. In particular, pairs such as (κ, θ) or (σ_v, ρ) can partially compensate for each other, generating a good aggregate fit with different economic readings. This implies that calibrated parameters should not be used uncritically as direct indicators of the "true" market dynamics. They describe, first of all, the risk-neutral surface observed in the specific time window and strike–maturity grid under study.

A useful analytical discipline is to separate qualitative interpretation from strong inference. It is reasonable to conclude, for instance, that a strongly negative estimated correlation is consistent with equity-like skew. It is much less reasonable to turn the exact numerical value of that correlation into a definitive statement about the deep structure of the market without further supporting evidence.

B.9 Conceptual ablation of the role of software

Because the report deliberately relegates software to a methodological support function, it is useful to ask which conclusions depend genuinely on the toolkit and which depend instead on the financial models themselves. This question can be read as a conceptual ablation.

If the toolkit were removed entirely, many theoretical results would remain intact: the FTAP, the logic of risk-neutral pricing, the binomial model, the Black–Scholes–Merton PDE, and the definition of the Greeks do not depend on software infrastructure. Some qualitative insights on discrete hedging and implied volatility could also be stated in purely theoretical language. What would disappear, however, is the ability to measure, visualise, and compare those insights systematically. One would no longer be able to show with the same force the convergence of Monte Carlo to the benchmark, the dispersion of hedging P&L, the morphology of the implied surface, or the geography of calibration residuals.

The ablation therefore suggests a clear conclusion: software does not create the theory, but it makes the quantitative argument testable and auditable. That is precisely why it remains in the report, albeit in a subordinate position. The scientific core of the project lies in the

questions about pricing, hedging, and calibration; the toolkit is the means by which those questions are turned into organised numerical evidence.

B.10 Residual limits and future directions

The supplementary analyses collected here also make the remaining limits of the present study more visible. The first concerns empirical scale: future work should ideally use denser market surfaces, possibly across multiple observation dates and with explicit treatment of bid–ask spreads and liquidity. The second concerns the integration of pricing and hedging: a natural research direction is to compare models not only by cross-sectional fit, but also by out-of-sample hedging quality. The third concerns model expansion: local volatility, jump models, mixtures, or rough volatility would make it possible to test whether a better fit justifies the additional numerical complexity.

Another promising direction is to make experimental reporting more systematic. Complete manifests of run parameters, standardised serialisation of results, automatic comparison across seeds, and comparative report generation could further strengthen the methodological value of the toolkit without changing its role as research support.

B.11 Appendix conclusion

The considerations in this appendix converge toward a simple idea: in quantitative finance, the robustness of a result depends not only on the model chosen, but also on the discipline with which sensitivities, trade-offs, and numerical limits are analysed. Ablation does not complicate the picture for its own sake; it makes the picture more credible. Showing how conclusions change with the number of paths, the seed, the loss function, the hedging frequency, and the density of data helps distinguish structural results from accidental ones. In that sense, the supplementary analyses strengthen the general claim of the project: sound theory is indispensable, but it acquires full value only when subjected to systematic numerical validation and interpreted with methodological caution.

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Note on AI-assisted translation

This document used generative AI exclusively as support for the translation of the text into English. The quantitative content, mathematical arguments, model discussion, numerical interpretation, and overall research structure were developed independently.